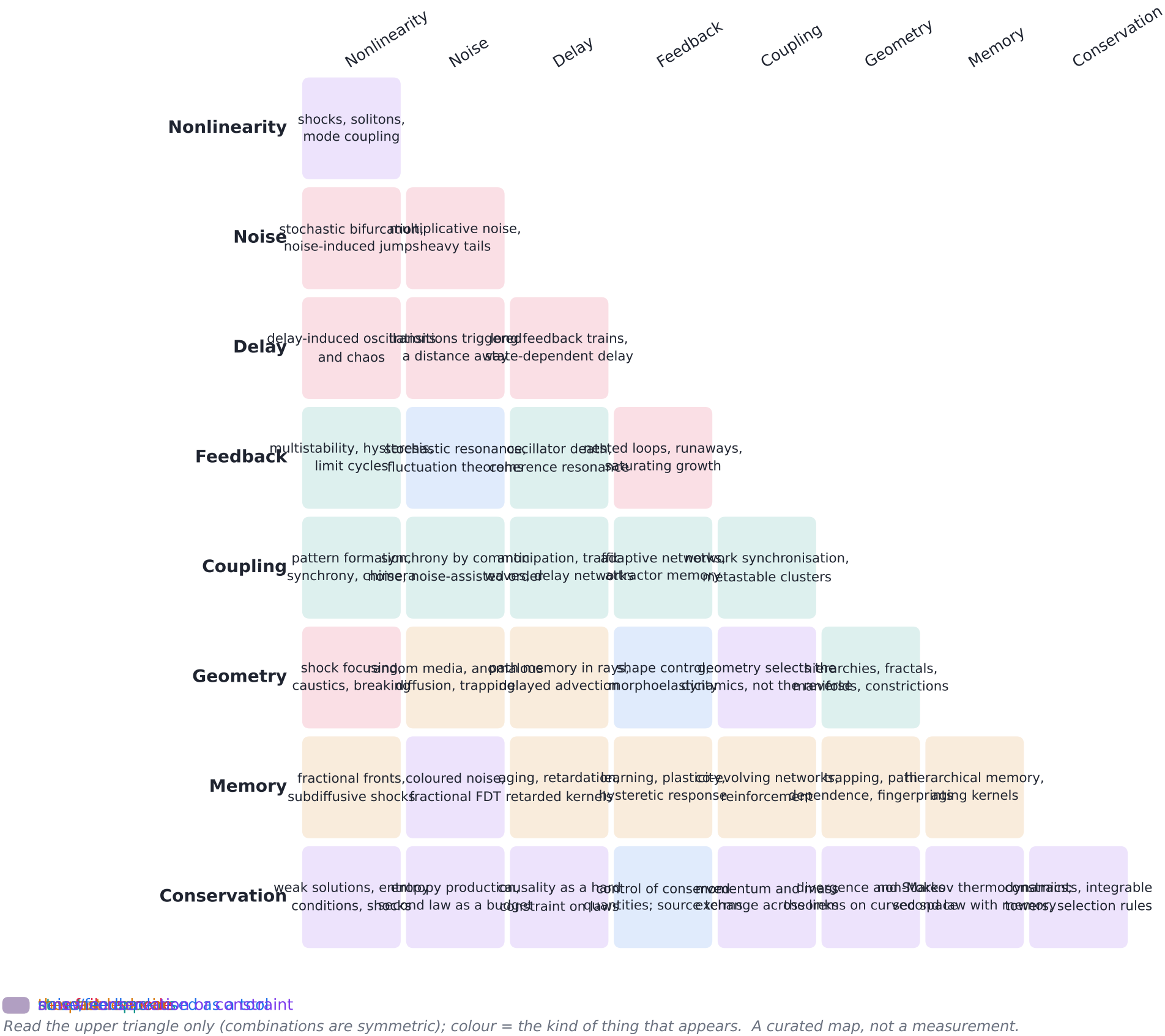
Nonlinearity
Noise
Delay
Feedback
Coupling
Geometry
Memory
Conservation
Nonlinearity
shocks, solitons, mode coupling
Noise
stochastic bifurcations, noise-induced jumps
multiplicative noise, heavy tails
Delay
delay-induced oscillations and chaos
transitions triggered at a distance away
long feedback trains, state-dependent delay
Feedback
multistability, hysteresis, limit cycles
stochastic resonance, fluctuation theorems
oscillator death, coherence resonance
activated loops, runaways, saturating growth
Coupling
pattern formation, synchrony, chimera
synchrony by common noise, noise-assisted
anticipation, traveling waves, delay networks
frustration, adaptive networks, attractor memory
network synchronisation, metastable clusters
Geometry
shock focusing, caustics, breaking
sound in random media, anomalous diffusion, trapping
path memory in rays, delayed advection
shape control, morphoelasticity
geometry selects the dynamics, not the manifold
hierarchies, fractals, constrictions
Memory
fractional fronts, subdiffusive shocks
coloured noise, fractional FDT
aging, retardation, retarded kernels
learning, plasticity, hysteretic response
co-evolving networks, reinforcement
trapping, path dependence, fingerprinting
hierarchical memory, aging kernels
Conservation
weak solutions, entropy conditions, shocks
entropy production, second law as a budget
causality as a hard constraint on
control of conserved quantities; source exchange
momentum and mass across links
divergence and curl on curved space
Markov thermodynamics, detailed balance with memory
constraints, integrable dynamics, selection rules
Nonlinearity
Noise
Delay
Feedback
Coupling
Geometry
Memory
Conservation as constraint
Read the upper triangle only (combinations are symmetric); colour = the kind of thing that appears. A curated map, not a measurement.